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24 May 1974
SUPERSEDING
MIS-11041
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MILITARY SPECIFICATION

TERPOLYMER, ETHYLENE PROPYLENE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of ethylene propylene terpolymer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 412	Tension Testing of Vulcanized Rubber
ASTM D 2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer, Test for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction (preshipment) sample. When required (see 6.2), a preproduction (preshipment) sample shall meet the requirements of this specification.

3.2 Storage. The terpolymer shall meet the requirements of this specification after indoor storage at ambient temperature and humidity for a period of 12 months. It shall be wrapped and scaled in polyethylene material.

3.3 Physical. Physical requirements of the material shall be in accordance with Table I.

3.4 Workmanship. The ethylene propylene terpolymer furnished under this specification shall be a product free from foreign materials. It shall be uniform in quality and manufactured in accordance with standard manufacturing procedures of the industry.

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Table I. Physical Requirements

Characteristic	Requirement Minimum	Method Paragraph
Vulcanization Control Test		
Tensile Strength, pounds per square inch	1725	4.7.1.5
Modulus at 300 percent Elongation, pounds per square inch	300	4.7.1.5
Elongation at Break, percent	600	4.7.1.5
Hardness Shore "A"	54 plus or minus 2	4.7.1.5
Alternate Vulcanization Control Test		
Modulus at 300 percent Elongation, pounds per square inch	1200	4.7.2.5
Tensile Strength, pounds per square inch	3000	4.7.2.5
Elongation at Break, percent	400	4.7.2.5
Hardness Shore "A"	65 plus or minus 5	4.7.2.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier

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may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. An inspection lot shall consist of material produced by use of one continuous operation employing not more than one lot of each ingredient and without change in formulation or process.

4.2.1 Inspection lot packaging. The supplier shall furnish to the Government certification of compliance to the requirements of 5.1.

4.3 Quality conformance sampling. Sampling for quality conformance testing will be in accordance with variables sampling plans as specified in MIL-STD-414, using the standard deviation method, variability unknown. Double specification limit with an acceptable quality level of 1.5 shall be used. An inspection lot represented by a container of material not conforming to the requirements of this specification will be subject to rejection.

4.4 Classification of inspection. The inspection of the terpolymer shall be classified as follows:

a Preproduction inspection.

b Quality conformance inspection.

4.5 Preproduction inspection. Preproduction inspection shall consist of all tests specified in 4.7.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of tensile strength, elongation at break, and Shore "A" hardness as specified in 4.7.1.5.

4.7 Test methods.

4.7.1 Vulcanization control test.

4.7.1.1 Apparatus.

a Mill with 6 inch diameter rolls and 14.5 inch working distance between guides.

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b Press, hydraulic, steam heated.

c Clicking machine.

d Oven, circulating air, 134 degrees celsius (C) (275 degrees Fahrenheit (F)).

e Analytical balance.

f Banbury mixer, or equivalent.

4.7.1.2 Ingredients. The following ingredients having chemical and physical properties as shown shall be used in the weight percentages as specified in Table II (see 6.5).

Table II. Chemical and Physical Properties

Material	Chemical and Physical Properties
Ethylene Propylene Terpolymer	per 3.3
Carbon Black High abrasion furnace black (HAF)	Heating Loss, % max @ 230 F = 1.0 Ash Content, % max @ 1025 F = 1.0 Iodine absorption, mg iodine/gm carbon = 70-85 325 mesh screen residue, % max = 0.1
Organic Peroxide 2,5 - Bis (Tert-Butyl Peroxy) -2,5 Di Methyl Hexane	Active peroxide, % min. = 50.0 Particle size, % through 100 mesh screen min. = 99.9
Material	**Weight Percent (1)**
Ethylene Propylene Terpolymer Carbon Black High abrasion furnace black (HAF) Organic Peroxide	75.47 22.64 1.89 100.00
(1) Weight percent tolerance is plus or minus 0.5 percent of individual percentages.	

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4.7.1.3 Preparation of masterbatch. Masterbatch shall be prepared in accordance with the following:

4.7.1.3.1 Roll mill preparation and mixing.

a Remove all foreign material from the roll mill and catch pan.

b Maintain a maximum mix weight of 30 pounds.

c Polymer Breakdown. Set the mill gap at 1/4 inch. Blend polymer and cut down two times from each side of the mill. Mix for two minutes total.

d All 1/3 of the HAF, continuously picking up the droppings and returning to the bit (gap).

e All the second 1/3 of the Philblack 0, mixing per step (d) above.

f Add the remaining HAF, mixing per step (d) above.

g Mix material alternately 5 minutes on fast roll and 5 minutes on slow roll. Cut down and return droppings to the bite continually. Mix for 20 minutes total.

h Sheet the masterbatch from the slow roll and cool.

4.7.1.3.2 Compounding. Compounding shall be accomplished in accordance with the following steps:

a Remove all foreign material from Banbury mixers and turn on cooling tower water to the mixer.

b Add the ethylene propylene terpolymer to the mixer and mix for 5-6 minutes.

c Add HAF in three increments. Lower ram and mix for 3-4 minutes between first and second and second and third additions. Add third increment on top of ram and mix 7-10 minutes.

d Move ram up and down several times to shake adhering HAF into mix.

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e Shut off cooling water, lower ram and mix until a temperature of 147 degrees C (300 degrees F) is reached. Do not exceed a temperature of 147 degrees C (300 degrees F) as indicated by thermocouple readout.

f Dump mix from Banbury and sheet on roll mill with cooling water running through mill. Entire mix should be sheeted at a thickness of 1/4 to 3/8 inch within 15 minutes after dumping from the Banbury. Masterbatch shall be cooled for a minimum of two hours before additional processing.

g Masterbatch which is not upgraded within 12 hours after preparation must be stored in polyethylene bags labeled with masterbatch number.

4.7.1.4 Masterbatch upgrading on differential speed mill. Upgrading of masterbatch shall be performed in accordance with the following steps:

a Add Organic Peroxide curative to masterbatch material. If there has been a delay between masterbatch processing and curative addition which has cooled the material to room ambient, mill masterbatch for 3 minutes before adding curative. Cut down alternating from left to right side of mill, returning droppings to the bite continuously. Mix on fast roll, for a total of 8 minutes mixing time.

b Move the material to slow roll and then remove for cooling.

c Place 5 pounds of material on an even speed mill band and set opening to 0.052 inch to 0.060 inch. Run the entire 5 pounds of material through the mill eight times. Remove and put the material on clean paper.

4.7.1.5 Procedure. The test shall be performed in accordance with the following steps: (1)

a Tensile Strength pounds per square inch. Test in accordance with ASTM D 412.

b Modulus at 300 percent elongation, percent. Test in accordance with ASTM D 412.

(1) The material shall be cured for 60 minutes at 160 degrees C (320 degrees F) plus or minus 5.5 degrees C (10 degrees F).

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c Elongation at Break, percent. Test in accordance with ASTM D 412.

d Hardness, Shore "A". Test in accordance with ASTM D 676.

4.7.2 Alternate vulcanization control test.

4.7.2.1 Apparatus.

a Mill, with 6 inch diameter rolls and 14.5 inch working distance between guides.

b Press, hydraulic, steam heated.

c Clicking machine.

d Oven circulating air, 134 degrees C (275 degrees F).

e Analytical balance.

4.7.2.2 Ingredients. (see Table III.)

Table III. Ingredients

Material	Base Recipe	3x Batch per 6 inch Diameter Mill
Ethylene propylene terpolymer	100	300 plus or minus 0.8 gram
Stearic acid, double-pressed	1	3 plus or minus 0.008 gram
Zinc oxide, dry a sufficient amount at 121 degrees C (250 degrees F) for 3 to 4 hours. Place in clean jars; seal and date.	5	15 plus or minus 0.04 gram
SAF black, repeat drying procedure used for zinc oxide	50	150 plus or minus 0.4 gram
Bias (Di methyl) Thio carbamoyl di sulfide (TMTD)	0.75	2.25 plus or minus 0.006 gram

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Table III. (Cont'd.)

Material	Base Recipe	3x Batch per 6 inch Diameter Mill
Tellurium Diethyldithio-carbamate	1.5	2.50 plus or minus 0.01 gram
Sulphur, Tube Brand, 30-10 325 mesh	1	3 plus or minus 0.08 gram

4.7.2.3 Compounding.

4.7.2.3.1 Pre-mix. Pre-mix shall be performed in accordance with the following steps:

a Just prior to compounding a batch, pre-mix the stearic acid and zinc oxide. Also pre-mix the TMTD, Tellurium Diethyldithiocarbamate (see 6.3) and sulphur.

b Moisture in the compounding ingredients, or moisture introduced into the compound during the mixing operation, will affect the cure and cause low modulus values.

c Adjust the roll clearance to 0.050 plus or minus 0.005 inch and the roll temperature at 29 degrees C (85 degrees F) plus or minus 5 degrees C (9 degrees F).

4.7.2.3.2 Mixing schedule (see Table IV.)

Table IV. Mixing Schedule

Time (minutes)	Operation
0	Band the ethylene propylene terpolymer on the front roll
1	Add stearic acid-zinc oxide mixture
2	Add all of the carbon black distributing it along the width of the band. Adjust the roll clearance to maintain a rolling bank.

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Table IV. (Cont'd.)

Time (minutes)	Operation
6	Transfer the batch to the back roll and add the TMTD-tellurium diethyldithiocarbamate-sulphur mixture. (1)
9.5	Adjust roll clearance to 0.060 inches so that the sheeted batch, after cooling, has a thickness of 0.085 to 0.095 inches. Return the batch to the mill and band it on the front mill. Remove sheet from mill. Place the sheeted batch on a clean, flat surface and allow to cool.

4.7.2.4 Vulcanizate preparation.

a After the batch has cooled, mark one slab for tensile test showing direction of grain. The slab shall be cured between 16 and 24 hours after mixing.

b Cure the specimens 1 hour at 150 degrees C (320 degrees F) in a press.

4.7.2.5 Procedure. The test shall be performed in accordance with the following steps: (2)

a Modulus at 300 percent elongation shall be tested in accordance with ASTM D 412.

b Tensile strength shall be tested in accordance with ASTM D 412.

(1) During the period between 6 and 9.5 minutes: Cut the batch 12 times alternating from side to side. Remove the batch from the back roll, adjust roll clearance to 0.025 to 0.030 inch and "Cigar" 6 times. Weigh the batch and record the weight. If the loss or gain exceeds 2.0 grams, discard the batch and repeat. If weight change is less than 2.0 grams, continue.

(2) Vulcanizates shall be held at room temperature for a minimum of 3 hours before testing.

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c Elongation at break shall be tested in accordance with ASTM D 412.

d Hardness, Shore "A" shall be tested in accordance with ASTM D 676.

5. PREPARATION FOR DELIVERY

5.1 Shipping containers. The product shall be shipped in commercial type containers that are suitably sealed to prevent contamination of the contents by moisture or foreign matter and shall have resealing features. Shipping containers shall be constructed to withstand normal transportation and handling abuse and shall be suitable for indoor storage for a period of 12 months minimum.

5.2 Marking. Shipping containers shall be legibly and durably marked in accordance with MIL-STD-129. Marking information shall include, but shall not necessarily be limited to, the following:

a Manufacturer's name or trademark and address.

b Manufacturer's lot designation.

c Name of material and quantity in shipment and date of packaging.

d Specification number.

e Purchase order number.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as a base material for rubber compounds.

6.2 Ordering data. Procurement documents should specify:

a Title, number, and date of this specification.

b Whether or not a preproduction sample is required (see 3.1).

6.3 Safety warning. Tellurium diethyldithiocarbamate is toxic. Avoid skin contact and inhalation of dust. Use in a well ventilated area.

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6.4 Alternate test method.

a Manufacturer may substitute the alternate vulcanization test method 4.7.2 for conformance to qualification and acceptance inspection required by the specification.

b Purchaser's acceptance of the material shall be in conformance with 4.7.1. Material failing this test method shall be subjected to test method 4.7.2. Material failing 4.7.1 and 4.7.2 shall be subject to rejection.

6.5 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-11041, dated 17 August 1965.

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